

Conditional borrower cash flows and liquidation close-out

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State and counterfactual. Original draw P is USD-equivalent DAI; a hypothetical local-currency debt starts at PE_0 , where E is local currency per USD. The effective annual local borrowing rate r gives $D(t) = P(E_0/E_t)(1+r)^t$. The comparator is the same borrower retaining the original collateral without borrowing. Draw proceeds are assumed held at par; profits, reinvestment, taxes and actual local-token trades are not observed.

Opening costs. At $t = 0$, an externally funded fee fP , opening conversion friction sP , and opening gas $g/2$ are paid. The fee is not financed. Base assumptions are $r = 20\%$, $f = 0.5\%$, $s = 0.3\%$ each conversion, and $g = \$40$ split equally between opening and voluntary closing. They are illustrative, not measured prices.

Mutually exclusive settlement routes

$$\begin{aligned} \text{Voluntary full repayment at } h : \quad & B^{\text{repay}}(h) = P - fP - sP - g/2 - D(h) - sD(h) - g/2, \\ \text{Sale at liquidation time } \tau : \quad & S = (1 - a)V_\tau, \quad R = \min(S, D(\tau)), \quad W = D(\tau) - R, \\ & \Pi = \min\{\lambda D(\tau), [S - D(\tau)]_+\}, \\ & H = [S - D(\tau)]_+ - \Pi, \quad B^{\text{liq}}(\tau) = P - fP - sP - g/2 - V_\tau + H. \end{aligned}$$

V_τ is the owned collateral mark, a an auction haircut, R debt recovery, W issuer/lender asset impairment, Π a penalty capped by surplus, and H collateral proceeds returned to the borrower. The liquidated branch has *no further borrower repayment* at h , no voluntary exit swap, and no voluntary closing gas. The shortfall is assigned to claimholders under a stated nonrecourse assumption; a recourse contract would need a different ledger.

Numerical reconciliation, excluding all opening costs and haircut.

At sale: $P = 100$, $D = 100$	V	R	Π	H	W	B^{liq}
Collateral surplus; $\lambda = 10\%$	180	100	10	70	0	-10
Collateral shortfall; $\lambda = 13\%$	60	60	0	0	40	40

For every sale, $R + W = D$ and $R + \Pi + H = S$. Borrower collateral opportunity cost is $V - H$, never $V - H$ *plus* a separate repayment. These identities are checked in the executable tests.

Empirical identification boundary. The data supply official monthly FX and static terminal collateral shocks, not a within-horizon price path, actual τ , auction price or probability weights. The reported collateral result is solely a terminal ratio $c_0 P(1 - q)/D(1)$ below a stipulated threshold. It is not an executed-liquidation probability or expected borrower outcome.